UMC 0.11um RFCMOS Al Advanced Enhancement P+/Nwell Junction Varactor SPICE Model Document

Version 0.2 Phase 1
2010/07/20

Table of Contents

List of Figures

List of Tables

1 Introduction

1.1 Revision history

Version	Date	Owner	Comment
0.2 P1	2010/07/20	Ho Cheng Hu	1. Version up to 0.2 based on Si data. 2. Add document.

1.2 General information

This document serves as a reference to the design works of products that will be manufactured by UMC using the following process.

> 0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor Al Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

> G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

UMC provides global models for their many benefits and the model accuracy is guaranteed by a quantitative report of model fitting error against the hardware data, which can be found in the following sections.

1.3 Flexible Backend Process Options

The standard backend process for UMC 0.11um RFCMOS technology is 1P8M. However, some RF designs may not require all 8 layers of metal. Therefore the flexible backend process options are also supported. These backend options are: 1P6M and 1P8M.

For RF models, only the devices that are backend-related devices need to take the flexible backend process options into account, either in model scaling range or model parameters. These devices are:

1. Planar MIM capacitors (model parameters)
2. Planar inductors (model parameters)
3. Stacked inductor (scaling range)
4. Pad (both scaling range and model parameters)

2 P+/Nwell Junction Varactor RF sub-circuit Model

2.1 Model usage

Model File Name:

Spectre: L110AE_VARDIOP_RF_V021.lib.scs , L110AE_VARDIOP_RF_V021.mdl.scs
Hspice: L110AE_VARDIOP_RF_V021.lib , L110AE_ VARDIOP_RF_V021.mdl
Eldo: L110AE_VARDIOP_RF_V021.lib.eldo , L110AE_VARDIOP_RF_V021.mdl.eldo

Sub-circuit Model Name & Nodes: vardiop_rf (pplus nwell psub)

Corner Section:

- min: P+/NW Junction Varactor with minimum capacitance value
- typ: P+/NW Junction Varactor with typical capacitance value
- max: P+/NW Junction Varactor with maximum capacitance value

Input Parameters:

- wp: P+ finger width in m
- l: P+ finger length in m
- nf: P+ finger number

Model Usage:

1. For Cadence Spectre

- Copy ModelFileName.lib.scs and ModelFileName.mdl.scs into your simulation directory.
- Add the following statement in the input netlist to the simulator for 90% dimension shrinkage.

SetOption1 options scale=0.9
include “./ModelFileName.lib.scs” section=CornerSection

, where CornerSection could be typ, min, max.

- Follow the example below to call the model

x1 (pplus nwell sub) vardiop_rf *l=10e-6 wp=0.5e-6 nf=12 m=8*

2. For Synopsys HSPICE

- Copy ModelFileName.lib and ModelFileName.mdl into your simulation directory.
- Add the following statement in the input netlist to the simulator for 90% dimension shrinkage.

.Option scale=0.9
.lib “./ModelFileName.lib” CornerSection

, where CornerSection could be typ, min, max.

- Follow the example below to call the model

x1 (pplus nwell sub) vardiop_rf *l=10e-6 wp=0.5e-6 nf=12 m=8*

3. For Mentor Graphics Eldo

- Copy ModelFileName.lib.eldo and ModelFileName.mdl. eldo into your simulation directory.
- Add the following statement in the input netlist to the simulator for 90% dimension shrinkage.

.Option scale=0.9

.lib “./ModelFileName.lib.eldo” CornerSection

, where CornerSection could be typ, min, max.

- Follow the example below to call the model

x1 (pplus nwell sub) vardiop_rf *l=10e-6 wp=0.5e-6 nf=12 m=8*

2.2 Scope

In this report, RF modeling results of P+/NW junction varactors are described.

1. Device Architecture and Layout

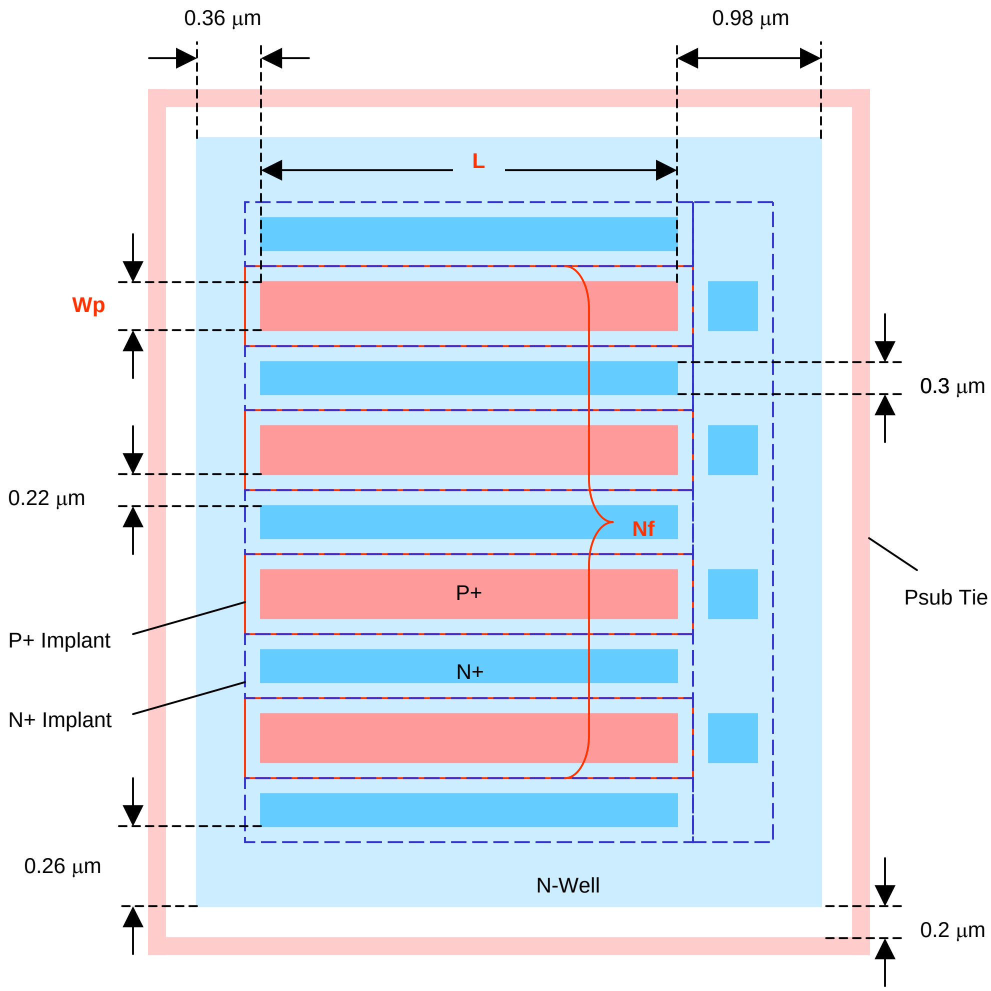

Figure 2-1 Example layout of P+/NW junction varactors

(All N+ and P+ diffusion areas are fully contacted.)

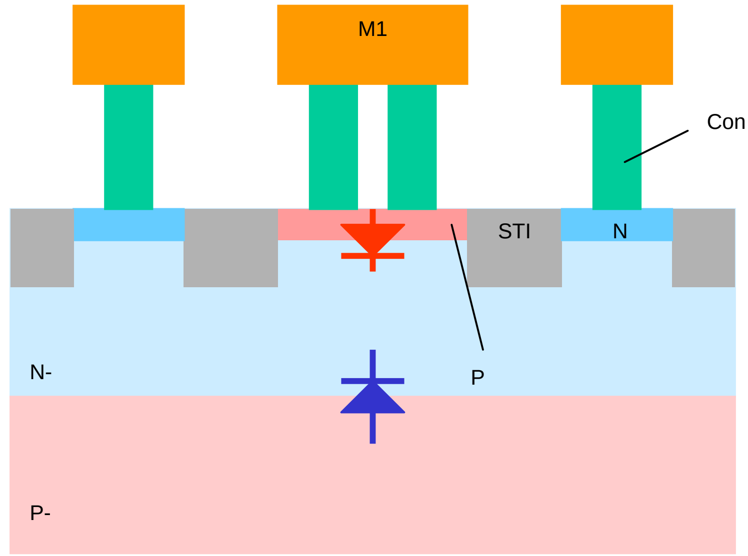

Figure 2-2 Cross-section of P+/NW junction varactors

2. Sub-circuit Model

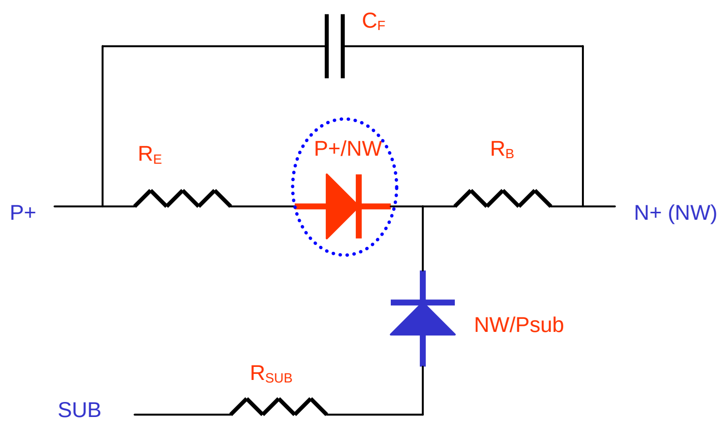

Figure 2-3 Sub-circuit model of P+/NW junction varactors

3. Model Equations

P+/NW diode:
$p\_area = l \cdot wp \cdot nf$
$p\_peri = 2 \cdot (l + wp) \cdot nf$

NW/Psub diode:
$nw\_area = [nf \cdot (wp + 0.44\mu) + (nf+1) \cdot 0.3\mu + 0.52\mu](l + 1.34\mu)$
$nw\_peri = 2[nf \cdot (wp + 0.44\mu) + (nf+1) \cdot 0.3\mu + l + 1.86\mu]$

re = re_unit/(l*wp*nf)
re_unit: P+ diffusion resistance per unit area

rb = rsh_nw*0.11μ/(l*nf)
rsh_nw: NW sheet resistance

cf = 2*cf_m1*l*nf
cf_m1: M1-M1 fringing capacitance per unit length

rsub = rsub_unit/nw_peri
rsub_unit: Substrate resistance per periphery

4. Corner Model

Component	Parameter	Min C	Max C
P+/NW	cjo	Nominal x 0.9	Nominal x 1.1
	cjsw	Nominal x 0.9	Nominal x 1.1
NW/Psub	cjo	Nominal x 0.9	Nominal x 1.1
	cjsw	Nominal x 0.9	Nominal x 1.1
Re	Re_unit	Nominal x 1.2	Nominal x 0.8
Rb	Rsh_nw	Nominal x 1.2	Nominal x 0.8
Cf	Cf_m1	Nominal x 0.8	Nominal x 1.2

5. Scaling Range and Device Usage Guideline

5.1 Recommended Scaling Range

Parameter	Unit	Min	Max
Wp	μm	0.28	2
L	μm	5	18
Nf	--	4	16
Freq	GHz	0.1	20.1
Vpn	V	-3.6	0.2

Temperature range is –55C to 125C

5.2 Device Usage Guideline

- Vpn (p+ to NW) should be always negative to avoid PNP turn-on.
- N+(NW) node is recommended as a control node.
- Model accuracy for the total series resistance (R_E + R_B) is not so good in general. This is partially due to variations in probe contact-resistance - this is one of issues to be improved in the final version model.

6. Model Accuracy Verification

6.1 Test Device Identification

- Lot No.: D4YAA
- Wafer No.: 7 (1P8M2T)
- Device ID: See Table 6.1

Table 2-1 P+/NW junction varactors used in this modeling

Device	Wp (um)	L (um)	Nf	M
PJV101	0.28	10	8	8
PJV102	0.5	10	8	8
PJV103	1	10	8	4
PJV104	2	10	8	2
PJV105	0.5	5	8	8
PJV106	0.5	18	8	4
PJV107	0.5	10	4	8
PJV108	0.5	10	16	4

6.2 Comparison of measurements and simulations

- In all measurements (denoted by .m), Port-1 and Port-2 of VNA were connected to P+ and N+(NW), respectively.

Table 2-2 Comparison of measurements and simulations @ Cj(0)(fF) at 5.8GHz

Device	Measurement	As-extract	Center
PJV101	401.1	397.7	379.6
PJV102	502.5	504.4	493.4
PJV103	375.5	373.5	376.1
PJV104	307.9	308.1	317.6
PJV105	257.3	256.1	250.0
PJV106	445.5	451.2	441.8
PJV107	250.1	252.7	247.2
PJV108	501.5	502.4	491.4

- Cj(0) is extracted in Vpn = 0V.

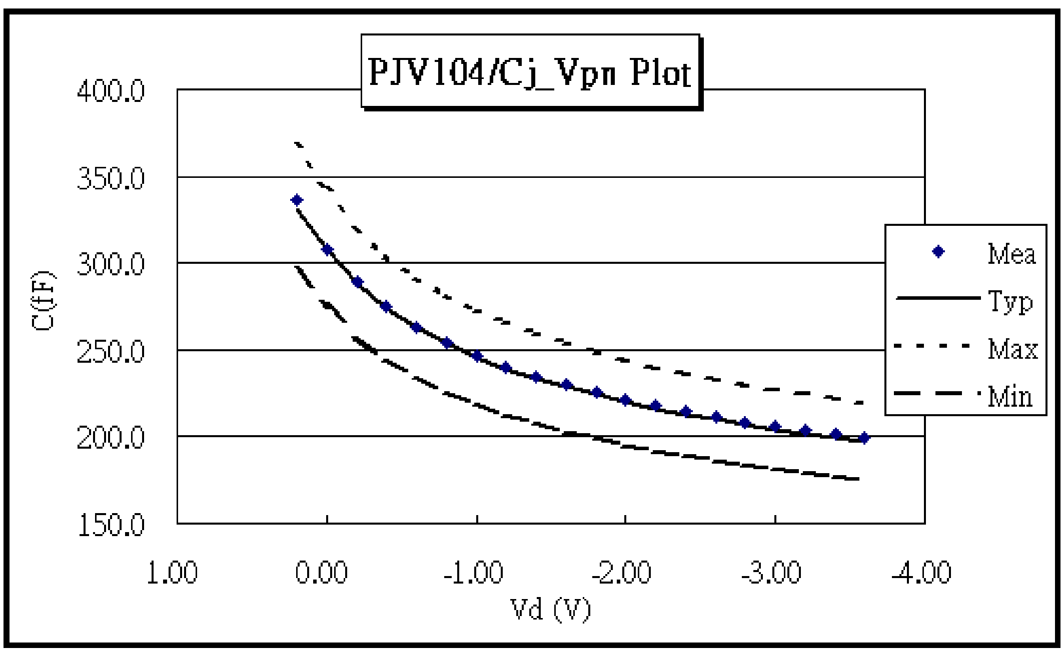

Figure 2-4 Capacitance vs Vpn for PJV104

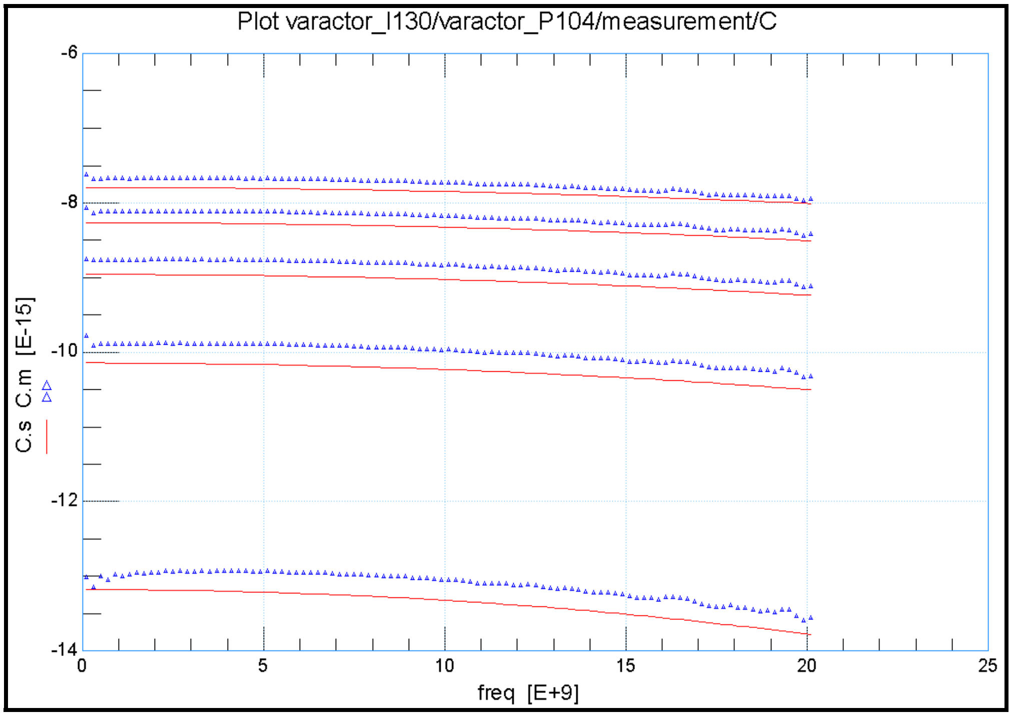

Figure 2-5 Capacitance vs Vpn for PJV104

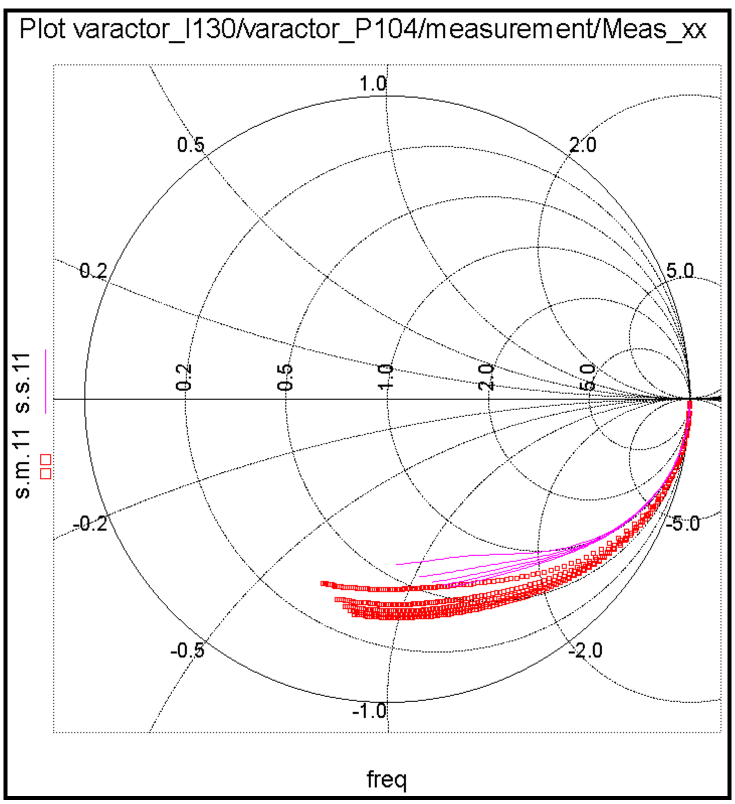

Figure 2-6 PJV104 S11 Plot

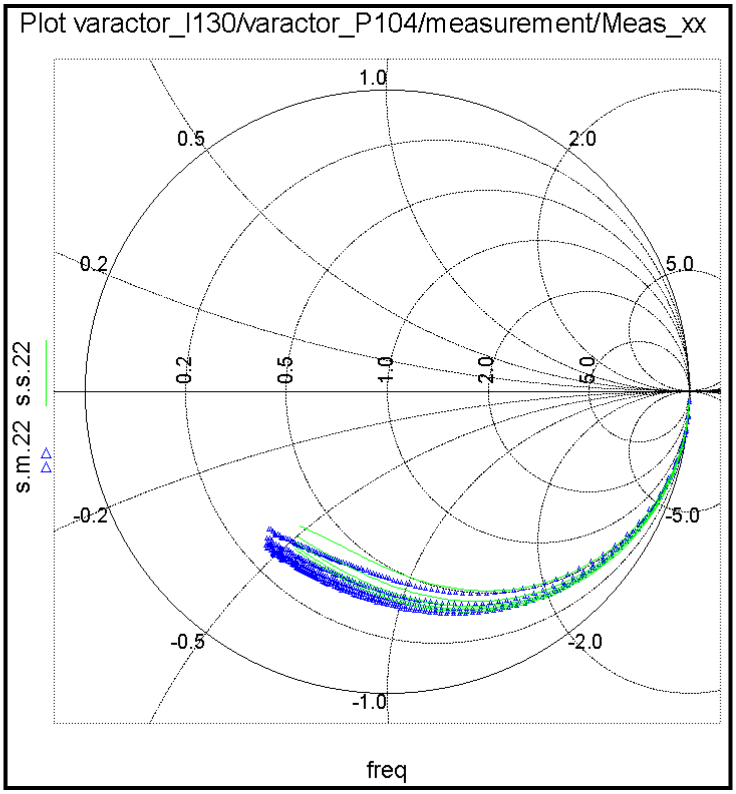

Figure 2-7 PJV104 S22 Plot

3 Appendix

3.1 HSPICE Warning Message

Table 3-1 HSPICE warning message

Item	Warning Message	Explanation
1	**None**	**None**

3.2 ELDO Warning Message

Table 3-2 ELDO warning message

Item	Warning Message	Explanation
1	**None**	**None**

3.3 SPECTRE Warning Message

Table 3-3 SPECTRE warning message

Item	Warning Message	Explanation
1	**None**	**None**